

Energy resilience: Solar and batteries are affordable options for U.S. households

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Standfirst

A majority of U.S. households can reduce energy costs and access affordable backup power during outages through rooftop solar and battery storage. Policymakers need to evaluate and adopt measures to ensure high-outage-risk and energy-burdened communities have equitable access to these adaptation solutions as climate impacts intensify outages.

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The policy problem

Climate extremes and aging grid infrastructure are increasing household energy burden and power outages across the U.S. Rooftop solar and battery storage, deployed together or separately, offer promising adaptation solutions as they may help to reduce electricity bills, provide backup power during outages, and lower carbon emissions. Policymakers can directly shape this landscape through tariff design, incentive programs, rate approvals, and industry regulations. However, a comprehensive understanding of where, for whom, and to what magnitude solar-battery systems provide cost savings or resilience benefits is needed, especially as various states are implementing tariff transitions such as from net metering to net billing. As climate impacts intensify, this knowledge gap undermines efforts to achieve affordable and equitable energy resilience for U.S. households.

The findings

Based on current market conditions, we find 60% of U.S. households can reduce their electricity costs by a mean 15%, and 63% can achieve affordable backup power covering 51% of essential energy needs

during outages on average, through solar-battery systems. These tangible benefits vary dramatically across states, however, and may not always align with areas of greatest need, particularly in high-outage-risk regions (See Fig. 1). The benefits may be less pronounced for rural and socio-economically disadvantaged communities, highlighting challenges with solar-battery adoption equality. Tariff analysis reveals transitioning from net metering to net billing has divergent outcomes: states like California and Florida maintain benefits while promoting battery adoption, but others like Iowa see substantial declines. Future technology cost reductions and rising retail rates could enhance benefits, although incentive phase-outs may limit gains.

The study

We analyze over 500,000 representative households from NREL's ResStock dataset across counties in the contiguous U.S. (contingent on physical suitability for rooftop installations). For each household, we first determine if installing solar and/or batteries could reduce electricity costs, and second, evaluate if such installations could provide affordable outage backup without increasing overall electricity costs. Each household's characteristics such as electricity consumption patterns are matched with location-specific data including technology costs, electricity rates, solar resources, and outage events. We focus on net billing, while comparing results to net metering scenarios. Using survival functions, we characterized the percentages of cost savings and backup coverage achievable nationwide and by state, enabling direct regional comparisons. Statistical modeling reveals which household and community characteristics influence access to these benefits. Our nationwide assessment enables direct policy comparisons, such as evaluating the transition between net metering and net billing tariffs.

Messages for Policy

- The majority of U.S. households can achieve cost savings or affordable backup with solar-battery systems under current market conditions. Public messaging on the benefits of solar-battery systems should therefore be expanded beyond early adopters to wider audiences.
- Lower solar-battery benefits in rural and disadvantaged communities risk worsening existing inequities in historical adoption patterns of these systems.
- Net billing's success in promoting battery adoption while maintaining solar-battery benefits in early states like California may not be replicable across all regions
- Policy incentives should target high-outage areas where current solar-battery backup benefits tend to be low, addressing the misalignment between system viability and household needs.

Further Reading

1. Shi, M., Lu, X. & Craig, M. T. Climate change will impact the value and optimal adoption of residential rooftop solar. *Nat. Clim. Change* 14, 482–489 (2024).

This article investigates how future climate change could increase the economic value of residential solar.

2. Forrester, S. P., Montañés, C. C., O'Shaughnessy, E. & Barbose, G. Modeling the potential effects of rooftop solar on household energy burden in the United States. *Nat Commun* 15, 4676 (2024).

This article explores how solar can reduce energy burdens for low income households.

3. Wussow, M. et al. Exploring the potential of non-residential solar to tackle energy injustice. *Nat Energy* 9, 654–663 (2024).

This article examines the potential of non-residential solar to benefit disadvantaged communities.

4. Yozwiak, M. et al. The effect of residential solar on energy insecurity among low- to moderate-income households. *Nat Energy* 10, 569–580 (2025).

This article shows that residential solar significantly reduces energy insecurity among low- to moderate-income households.

5. Gorman, W. et al. County-level assessment of behind-the-meter solar and storage to mitigate long duration power interruptions for residential customers. *Applied Energy* 342, 121166 (2023).

This article evaluates the technical capability of using residential solar-plus-storage systems for backup power across US counties.

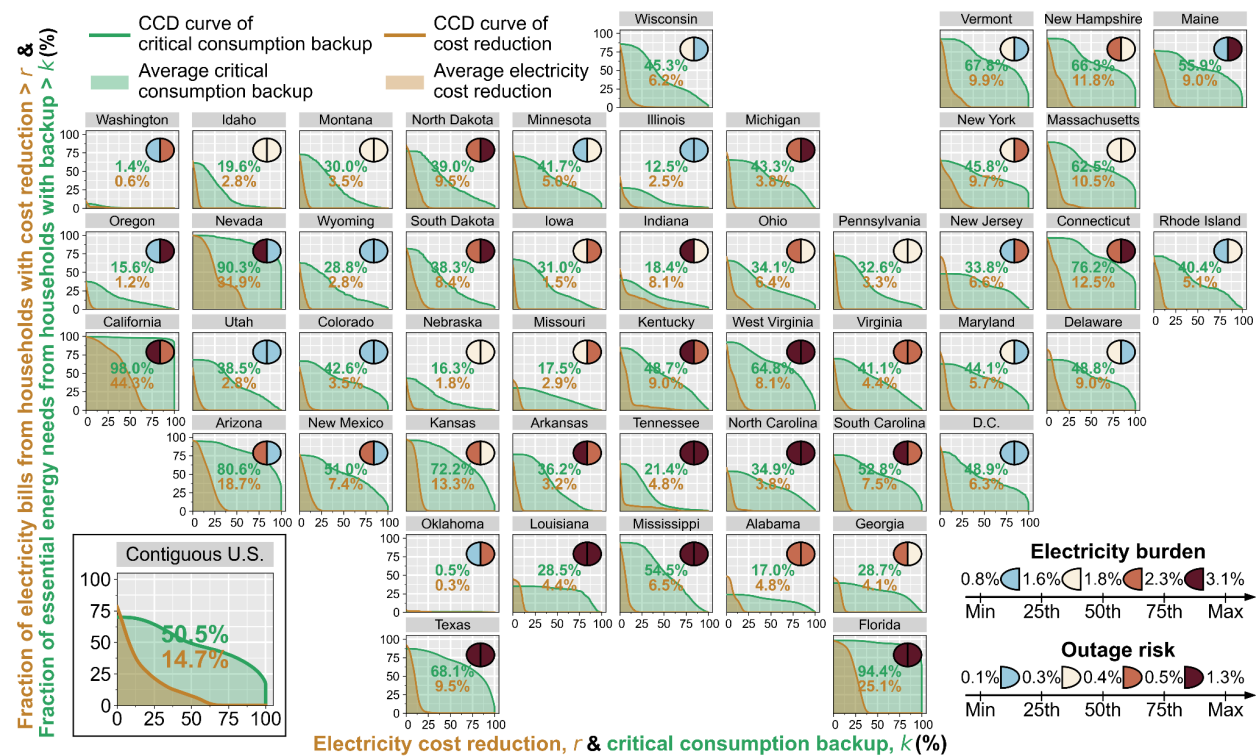


Fig. 1 | State-level solar-battery benefits under net billing tariffs, with electricity burden and outage risk context. For each state, two complementary cumulative distribution (CCD) curves, also known as survival functions, show distributions of household electricity cost savings (bronze) and backup power coverage (green). The shaded areas beneath these curves (marked as bronze/green text) represent state averages. Semicircles indicate each state's ranking among all states for electricity burden and outage risk. The national panel combines results from all contiguous U.S. states. Figure is adapted with permission from *Nat. Energy* [PRODUCTION PLEASE ADD DOI]; 2025.

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Competing interests

The authors declare no competing interests.